

Year 3 Science | Water Cycle Introduction

Teacher / parent map | Teach the gaps in order

Year 4 content preview: AC9S4U02 | Year 3 inquiry practice

Curriculum accuracy first

This is a Year 3-friendly introduction and Year 4 preview. The direct water-cycle content code is Year 4 AC9S4U02, not Year 3 AC9S3U01. Current Year 3 AC9S3U01 concerns living things and life cycles. Do not market this pack as required Year 3 water-cycle content.

What children will learn

Identify water stores; explain evaporation, condensation and precipitation; read arrow directions; connect processes to everyday evidence; record observations; explain one possible water journey. Runoff and soaking into soil add context. Detailed groundwater, transpiration and treatment systems are optional extensions.

Suggested sequence

Session 1: prior ideas and vocabulary (pages 2-3).

Session 2: diagram, labels and misconceptions (pages 4-6).

Session 3: supervised observation and recording (pages 7-8).

Session 4: routes, local application and independent check (pages 9-10).

Adult answers and review: pages 11-12. Adjust pace; do not assign all pages in one sitting.

Year 3 inquiry connections - selected aspects

AC9S3I01: questions and observation-based predictions (page 7).

AC9S3I04: diagrams and an observation table (pages 5 and 8).

AC9S3I06: communicating an explanation for an audience (page 9). These activities do not cover every aspect of each descriptor.

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Before we learn | Where is water?

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Name: _____ Date: _____

Draw or list three places where water can be found.

Think about the sky, land and ocean. Your drawings may include a river, puddle, cloud, lake or sea.

A puddle is smaller the next day. Where might its water go?

Write your first idea. You can change it after learning.

Show what you think

Circle one answer for each statement.

1. A cloud contains: invisible gas only / tiny droplets or ice crystals.
2. Rain falls from: clouds / the Sun.
3. A puddle can dry without boiling: yes / no.

My question about water

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Learn the words | Water changes and moves

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Water stores

Places where water is held: oceans, lakes, rivers, ice and water in the ground.

Evaporation

Liquid water changes into invisible water vapour, a gas. It can happen without boiling. Heat from the Sun helps power evaporation in nature.

Condensation

Water vapour cools and changes into tiny liquid droplets. Clouds can contain droplets and ice crystals; they are not just invisible water vapour.

Precipitation

Water falls from clouds, for example as rain, snow or hail. Droplets or ice particles grow large enough to fall.

Runoff and soaking in

Runoff flows over land towards streams, rivers or other lower places. Some water soaks into soil. Water may be stored before moving again.

Tell a partner: which processes change the form of water?

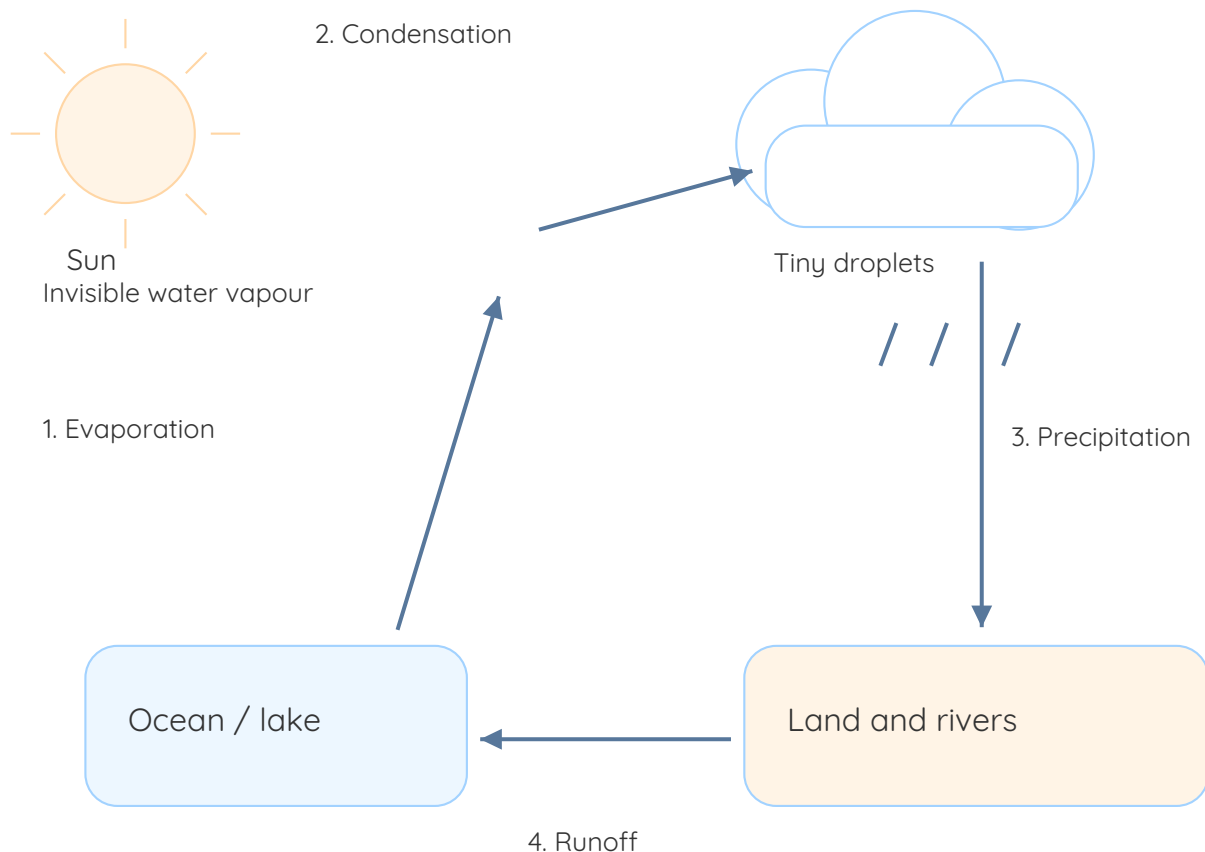
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A worked diagram | Follow one possible route

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Read the arrows, not just the labels.

The Sun supplies energy; gravity helps water fall and flow.



Water can also soak into soil or stay in lakes. This shows one possible route, not every route.

Explain the route

1. Water leaves the lake as invisible gas.
 2. Vapour cools and forms droplets.
 3. Water falls from the cloud.
 4. Some water flows over land back towards a lake or ocean.
- There is no single starting point. Water does not follow exactly the same route every time.

Model limit: arrows show changes and movement; vapour is invisible.

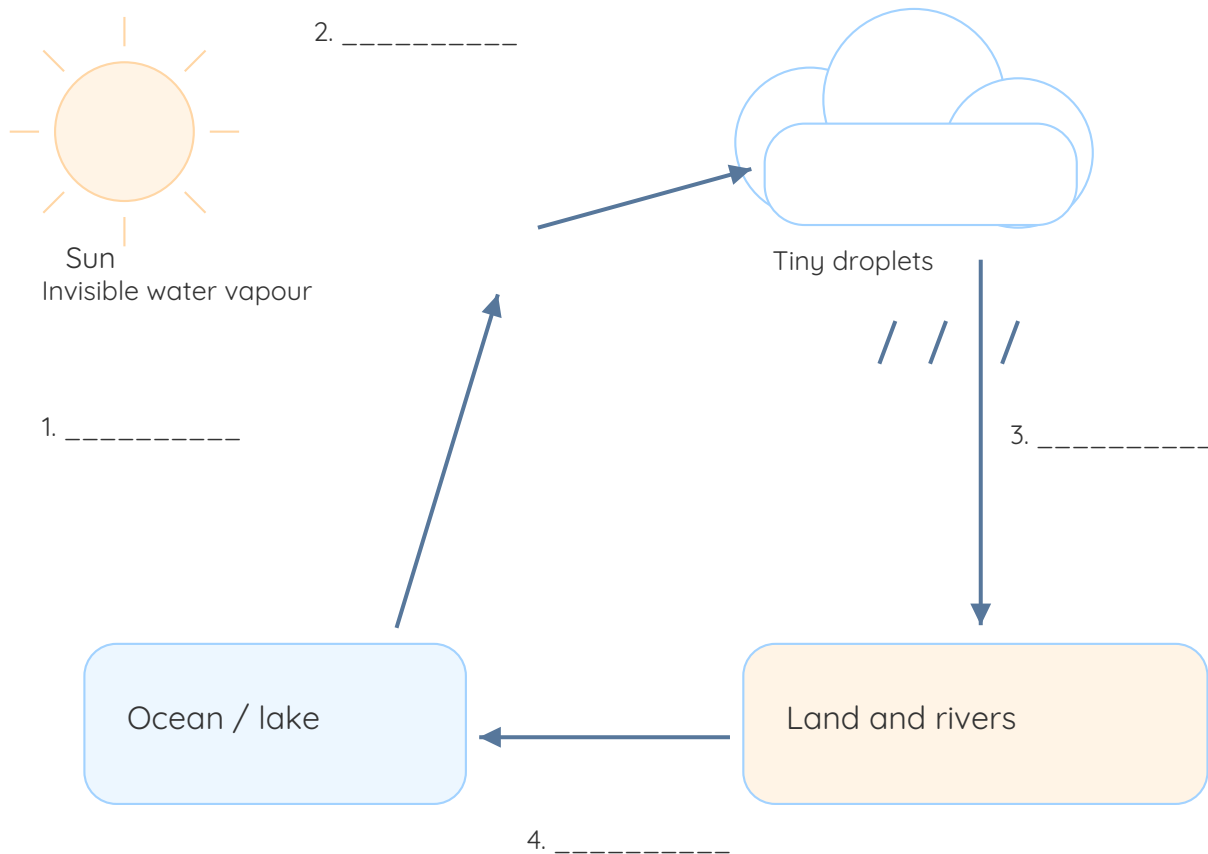
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Your turn | Label and explain

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Use the word bank to label arrows 1-4.

evaporation | condensation | precipitation | runoff



Water can also soak into soil or stay in lakes. This shows one possible route, not every route.

What change happens at arrow 1?

What change happens at arrow 2?

Tell a partner: could rain soak into soil instead of reaching the ocean?

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Science detective | Explain everyday evidence

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1. The disappearing puddle

A puddle dries on a warm day. Sam says, "The water has been destroyed."

Explain a better idea. Where could the water be now?

2. The wet outside of a cold cup

A sealed, dry cup holding cold water develops droplets outside. Mia says, "The water came through the cup wall."

Where did these droplets most likely come from?

3. The cloud question

Leo says, "A cloud is made only of water vapour."

What is wrong with this idea? Use the words gas and droplets.

4. The one-route mistake

A diagram shows rain flowing into a river. Does all rain travel straight to a river?

Give one other place it could go or stay.

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Investigate | Where do outside droplets come from?

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Question and prediction

Question: What difference do we observe on the outside of a cold container and a room-temperature container?

My prediction, based on something I have seen: _____

Materials and safety

Two identical watertight plastic containers with lids, ice water, room-temperature water, tray, cloth and timer. Adult supervision. No hot water, glass or tasting. Wipe spills. Wash hands afterwards.

Follow this procedure

1. An adult prepares one container with ice water and one with room-temperature water. Keep fill levels similar and fit the lids securely.
2. Wipe both outsides completely dry. Label Cold and Room temperature.
3. Put them side by side on a dry tray, away from splashes.
4. Observe the outside at 0, 5 and 10 minutes. Use page 8.
5. Describe differences. Do not wipe again during the observation.

Plan what stays the same

Container type, location, observation times and dry starting surfaces.

Main difference: cold contents versus room-temperature contents.

A question I could investigate next

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Record observations | Evidence before explanations

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Use your real observations.

Write dry / few droplets / many droplets, or describe what you see. No visible droplets is a valid result. Do not fill in what you think should happen.

Time	Cold: outside surface	Room temp: outside
0 min		
5 min		
10 min		

What difference or pattern did you observe?

Explain your evidence to a partner.

I observed _____

This is consistent with condensation because _____

If droplets did not form, what could we check or try next?

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A water journey | More than one possible route

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Tell a younger child how water can move.

Start with water in a lake. Explain evaporation, condensation and precipitation. Then choose one destination for the rain: a river, a puddle, soil or another water store. Use arrows or sentences.

My explanation

Make one local connection

Name a local water source, if you know one. Ask an adult or use a local water-provider source to check it. Do not assume all tap water comes straight from the nearest river.

Care for water

Suggest one way to reduce water waste. Why is saving usable water sensible even though water cycles?

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Independent check | Hide the teaching pages

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1. Explain the changes.

- a. What is evaporation? _____

- b. What is condensation? _____

2. Name the process.

- a. Rain falls from a cloud: _____
- b. Water flows down a slope towards a creek: _____

3. Explain a new example.

A wet shirt dries on a washing line. What happens to some of its liquid water? Must the shirt become boiling hot?

4. Draw one possible route.

Include a water store and label three main processes. Use arrows.

My next step

☐ I can explain changes. ☐ I need another example.

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Answer guide | Accept scientifically sound wording

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Pages 2-5

Page 2: water stores may include sea, river, lake, soil water or ice. Clouds contain droplets and/or ice crystals; rain falls from clouds; puddles can dry without boiling. Treat initial answers as diagnostic.

Page 5: 1 evaporation; 2 condensation; 3 precipitation; 4 runoff. Arrow 1: liquid to gas. Arrow 2: gas to liquid. Rain can soak into soil.

Page 6: misconceptions

1. Water may evaporate into air; it is not destroyed. A puddle may also soak into ground.
2. Outside droplets most likely form from vapour in surrounding air cooling at the cold surface, not water passing through an intact wall.
3. Vapour is an invisible gas; clouds contain tiny droplets and/or ice crystals.
4. Rain can soak into soil or remain in puddles, lakes or other stores.

Pages 7-9: investigation and explanation

Predictions vary; ask for an observation-based reason. Cold surfaces often collect more droplets, but humidity and surface temperature affect results. No droplets does not disprove condensation. Check for spills or leaks before interpreting wetness.

Journey explanations must describe changes, not just name stages. Water-saving examples may include repairing leaks or turning taps off when not needed. Fresh usable water is not always available where and when needed.

Page 10: independent check

1a liquid water becomes gas; 1b water vapour becomes liquid droplets.

2a precipitation; 2b runoff.

3 water evaporates into air; boiling is not required.

4 accept any plausible labelled route with evaporation, condensation and precipitation. Arrows must show appropriate movement.

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Adult guide | Differentiate, assess and audit

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Support / core / challenge

Support: read aloud; accept oral responses and labelled drawings; rehearse “liquid to gas” and “gas to liquid”; supply vocabulary cards.

Core: complete labels and explain changes in sentences.

Challenge: add a branching route and explain a model limit. Investigate local water sources using a reliable local source.

Observe understanding

Record Not yet / With support / Independently for: identifies a water store; explains evaporation; explains condensation; identifies precipitation; uses arrows; links observation to explanation. Note adult help. Reteach any process the child only names but cannot explain.

Investigation limits and next teaching steps

This comparison models condensation on a surface, not the whole water cycle or cloud formation. It is not a complete formal fair-test assessment. In dry air there may be little visible moisture. An adult can repeat under different conditions. Optional follow-up: mark a shallow puddle on a non-porous tray and observe evaporation over time.

Create-Audit-Improve / War Room checks

Corrected the year/code mismatch; separated invisible vapour from visible droplets; added branching routes, genuine observation records and independent transfer. Resource quality checked; classroom impact and sales demand are untested.

Primary references | Checked 8 October 2026

ACARA: AC9S4U02 teacher background information.

Scoutle: AC9S3I01, AC9S3I04, AC9S3I06 current v9 descriptions.

USGS Water Science School: Evaporation and the Water Cycle; Condensation and the Water Cycle.

Full source links and page-to-skill map accompany this resource.